

# Introduction to Deep Learning for Life Sciences

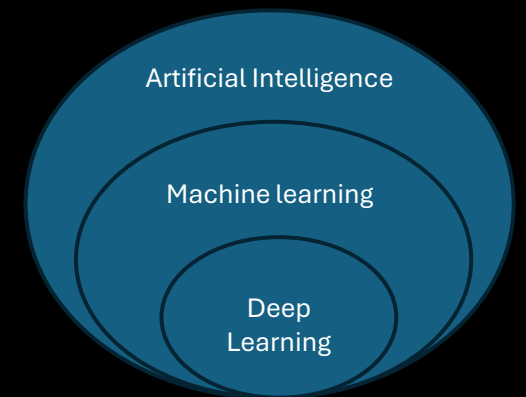
Nuno Pimpão Martins

Carvalho-Santos lab @ GIMM and Henriques lab @ ITQB-Nova



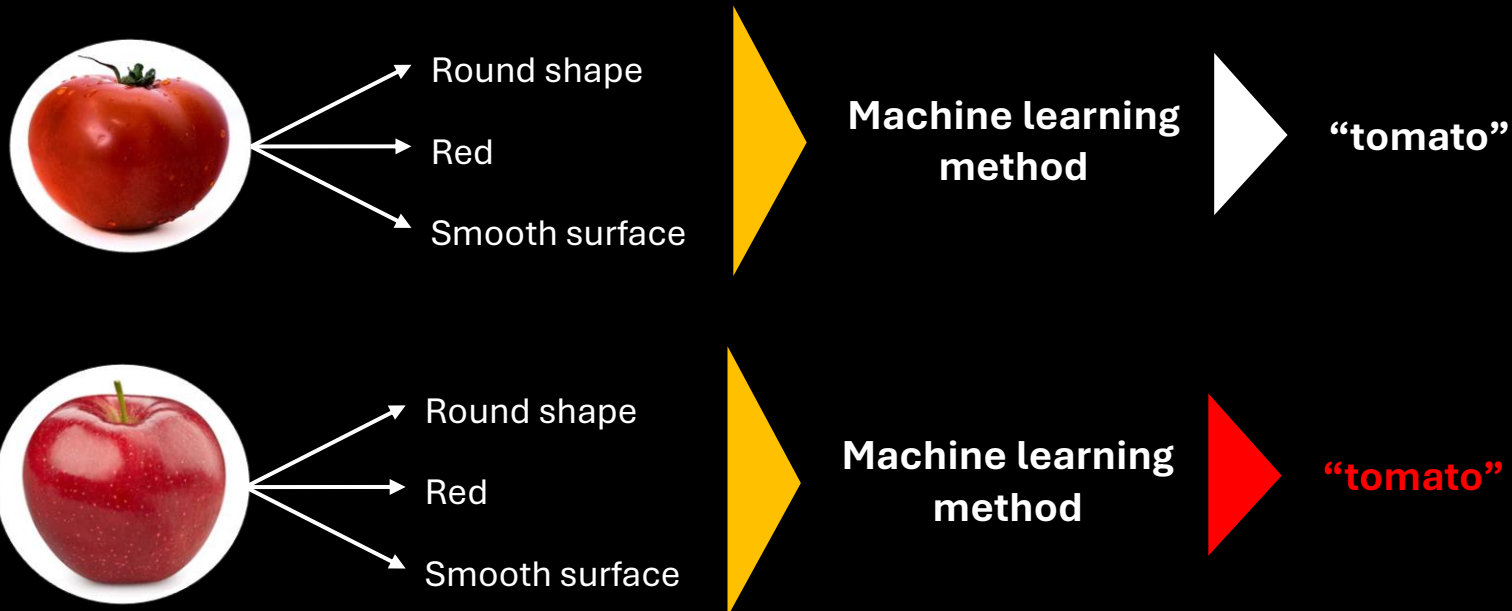
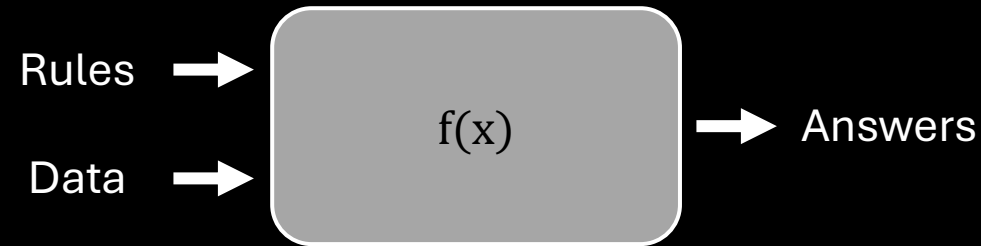
# Deep Learning (DL) is a subset of Machine Learning algorithms

- Artificial Intelligence (AI) – term invented in the 1950s: “(...) *the effort to automate intellectual tasks normally performed by humans* (...)”
- Are expected to have some level of “reasoning”
- DL and Artificial Neural Networks have been around for many decades.
  - 1943: *Perceptron*, the “first” artificial neural network, by Warren McCulloch and Walter Pitts
  - 1960s: Deep Neural Networks first appeared in the 1960s
  - 1986: modern backpropagation algorithm by Geoffrey Hinton (Nobel and Turing award laureate)
  - 1980s: Convolution Neural Network (CNN) are used for image recognition (LeNet by Yann LeCun and others)
  - 2010s: start of DL boom (AlexNet and ImageNet challenge)



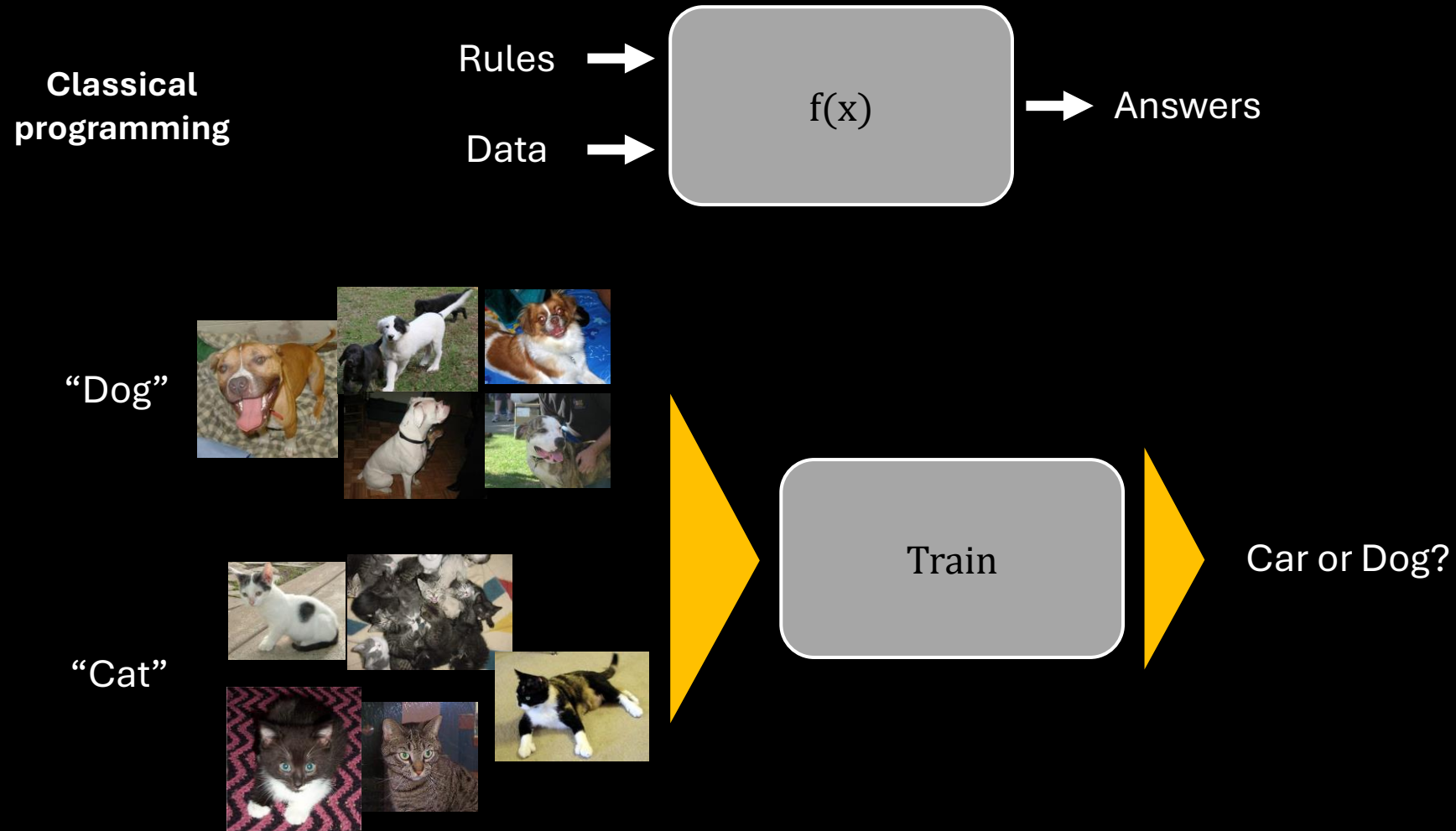
# Shift in computational paradigm

**Classical programming**



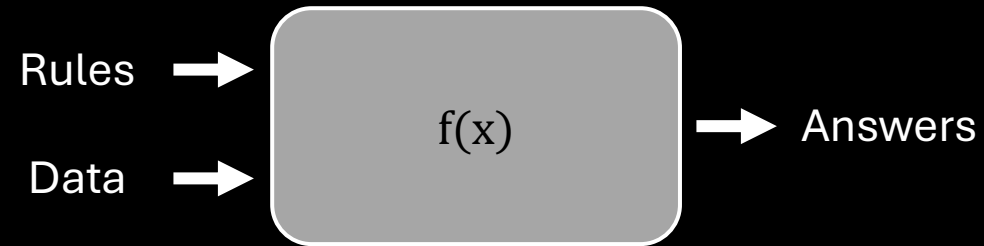
What if the algorithm learns from data automatically?

# Shift in computational paradigm

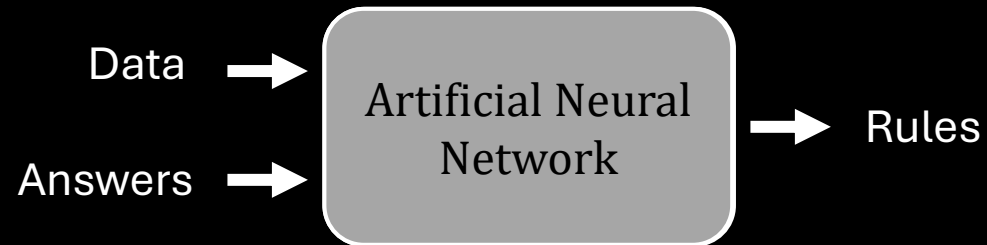


# Shift in computational paradigm

**Classical  
programming**

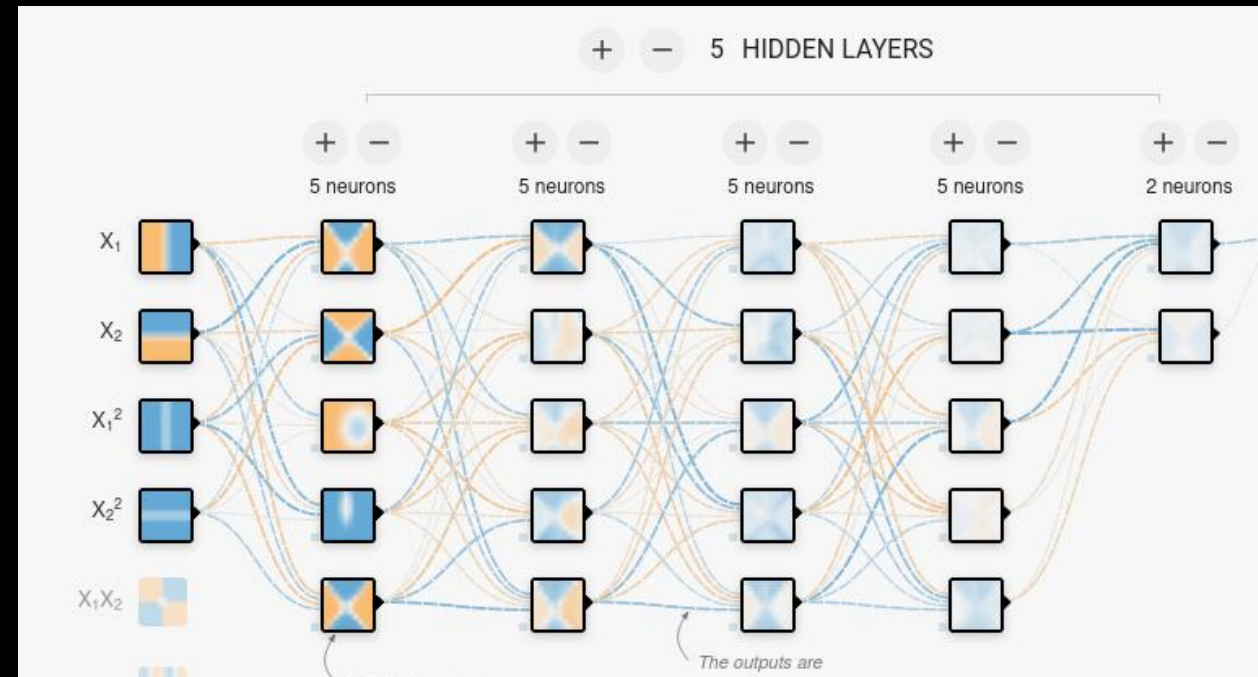


**Deep  
Learning**



# What are Artificial Neural Networks?

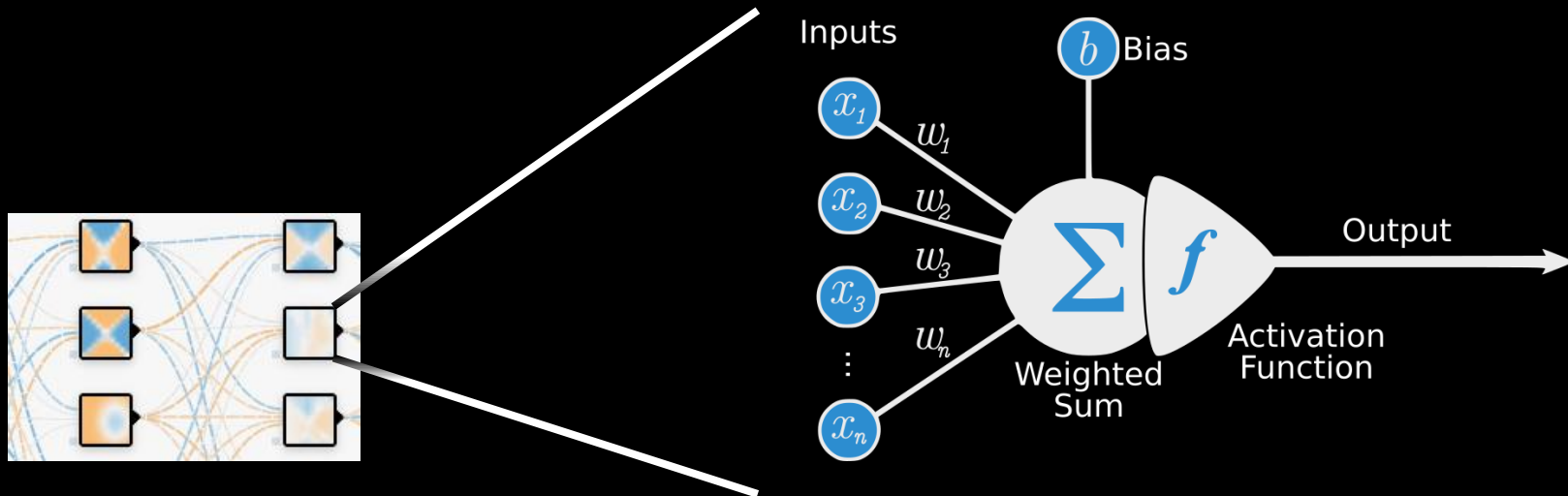
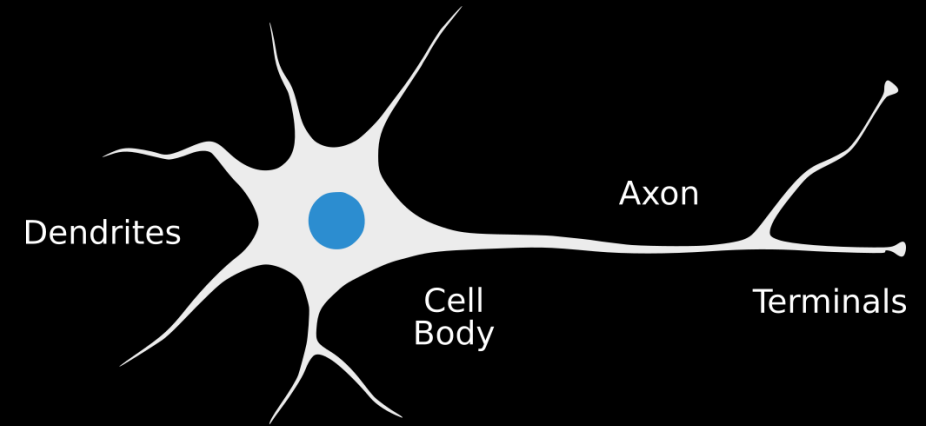
- Artificial Neural Networks (ANN) were inspired by neuronal connections
- Network of connected layers (vary depending on network architecture)
- Each layer is composed by many neurons
  - Nodes on the layer
- Deep Learning networks are characterized by having many “hidden” layers



Tensorflow Playground (<https://playground.tensorflow.org>)

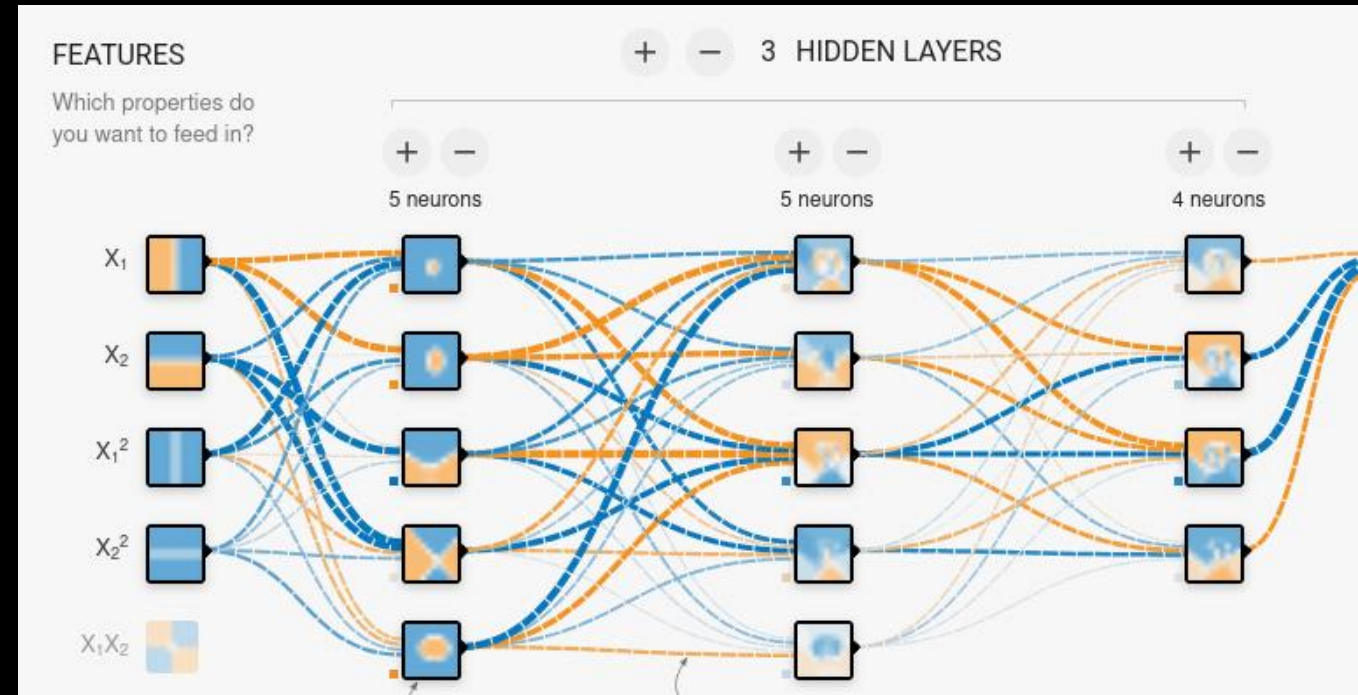
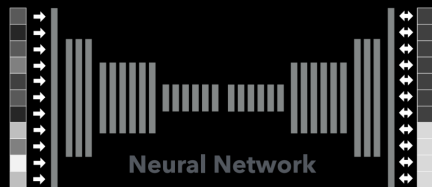
# What are Artificial Neural Networks?

- Input from previous nodes are combined and summed (with weights)
- Each network node passes information when activated
  - Signal goes over a certain threshold



# What are Artificial Neural Networks?

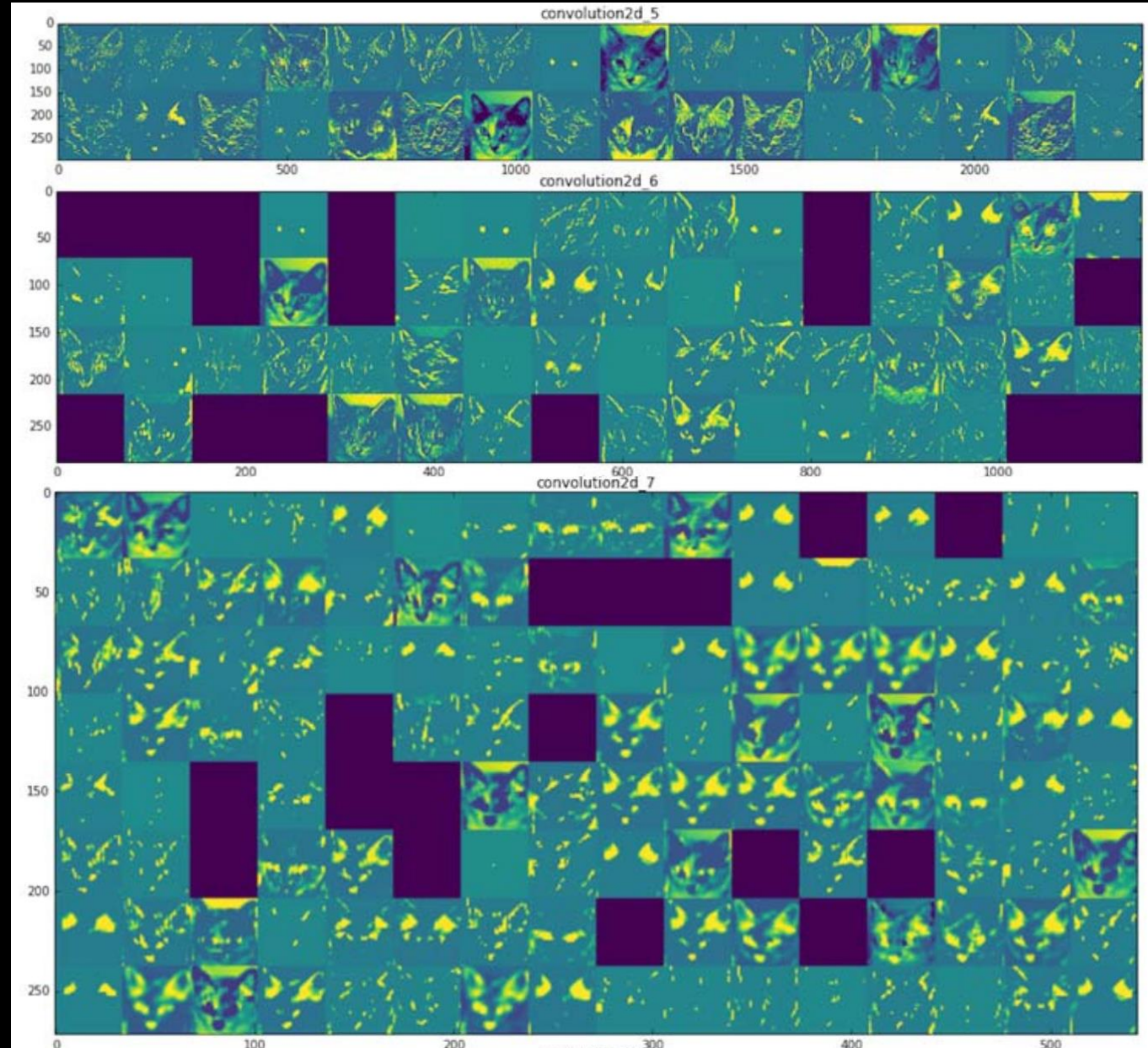
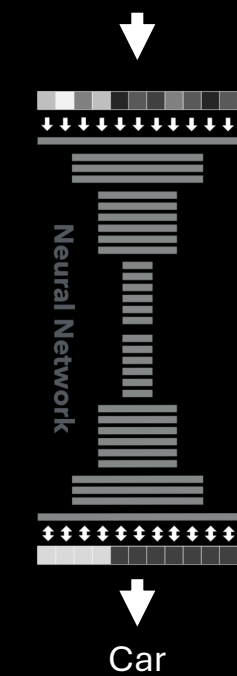
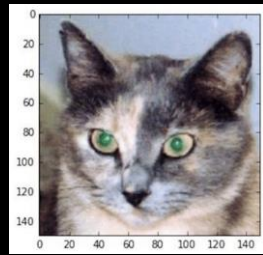
- Input from previous nodes are combined and summed (with weights)
- Each network node passes information when activated
  - Signal goes over a certain threshold
- Different image patterns activate different subsets of neurons
- Most computer vision and image processing ANN are Convolutional Neural Networks (CNN)



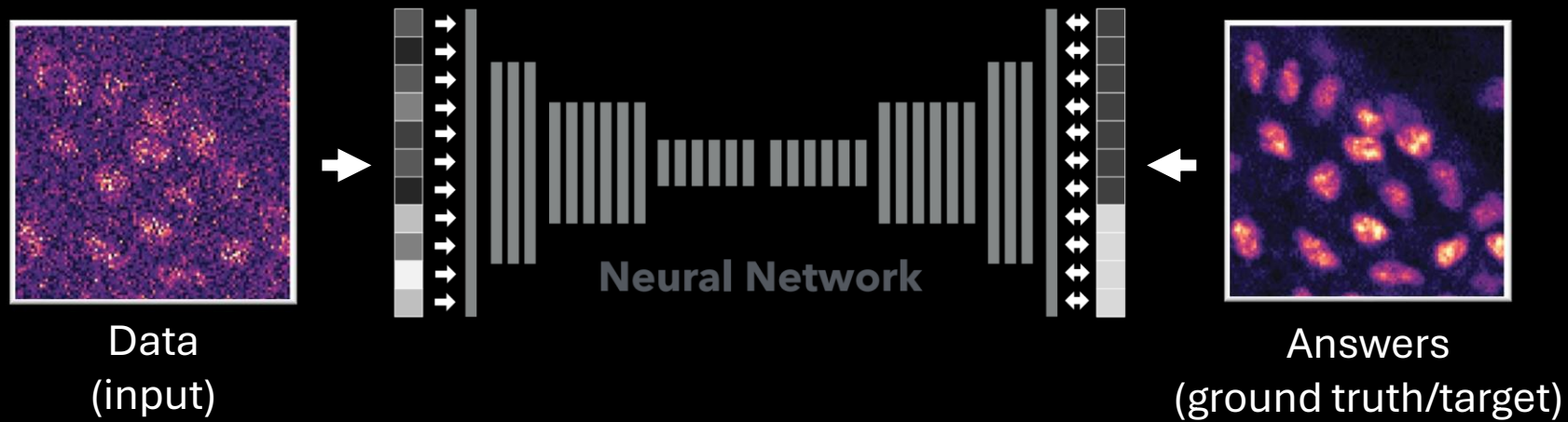
Tensorflow Playground (<https://playground.tensorflow.org>)



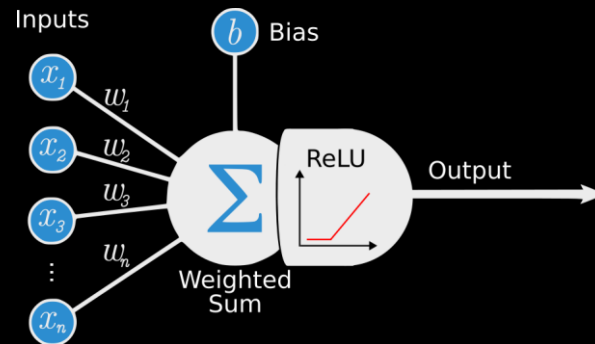
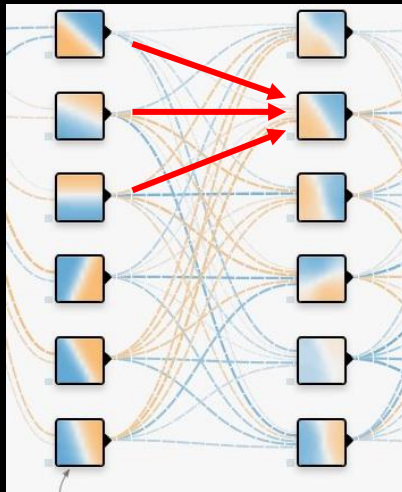
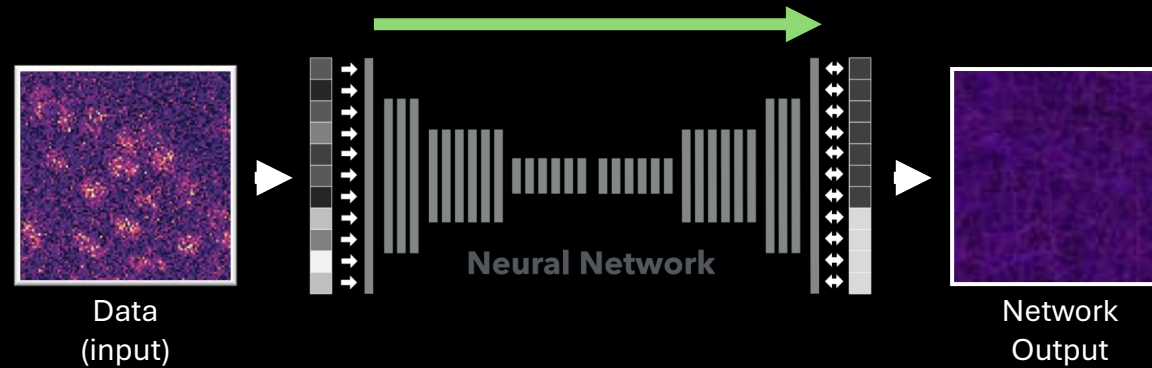
# ANN can generate complex filters which are more “abstract”



# How do you train a DL network/model?

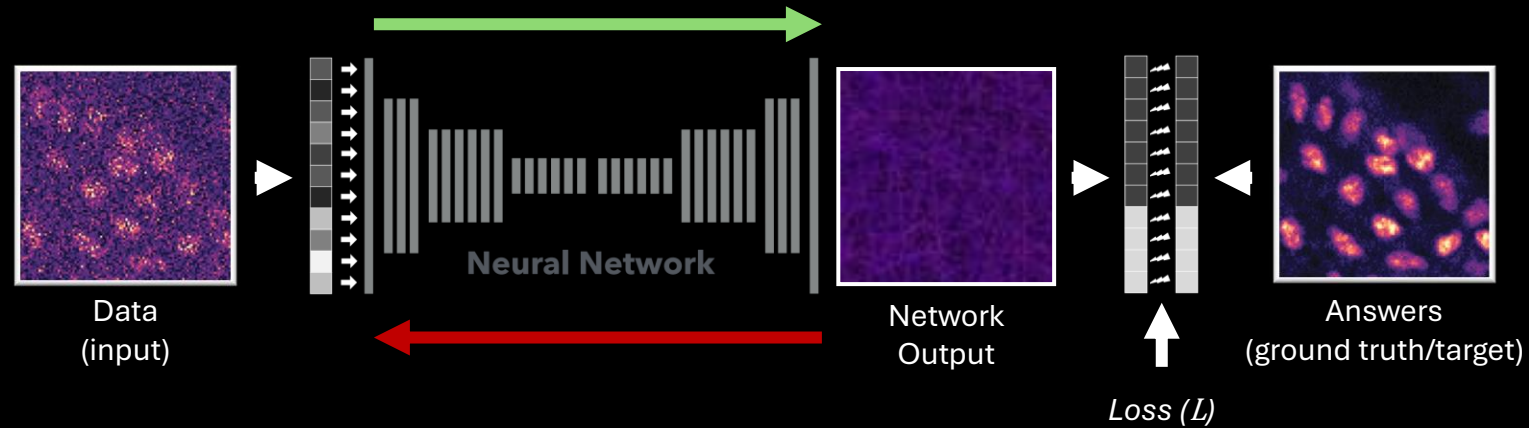


# Forward pass sets values and weights with node activations



- Each node input is multiplied by a weight
- All weighted inputs are summed
- Pass through activation function
- If: output > activation threshold → signal passed on
  - ReLU (rectified linear unit) is a common activation function

# Training loss tells the network how wrong it is and how to adjust

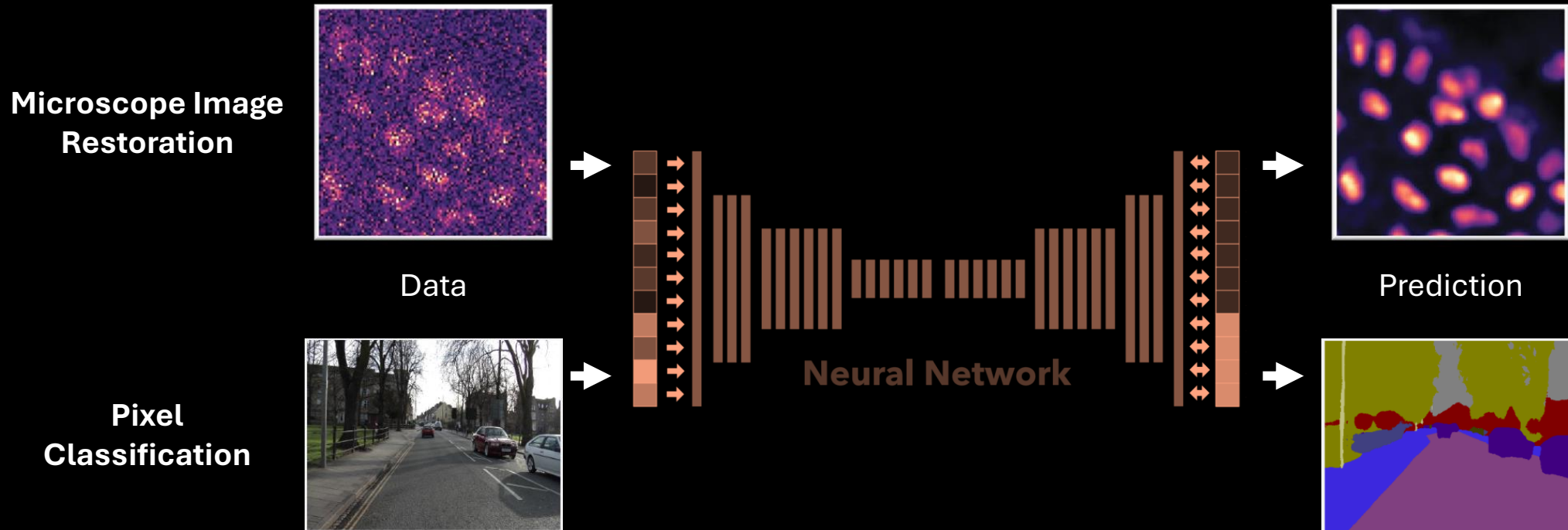


- Network output is compared to ground truth data
- A *Training Loss* (or *error*) is measured
  - Metric that measures difference between output and truth
- Error is backpropagated through the network
- Rinse and repeat (the training part of a network)
  - Over many steps and epochs



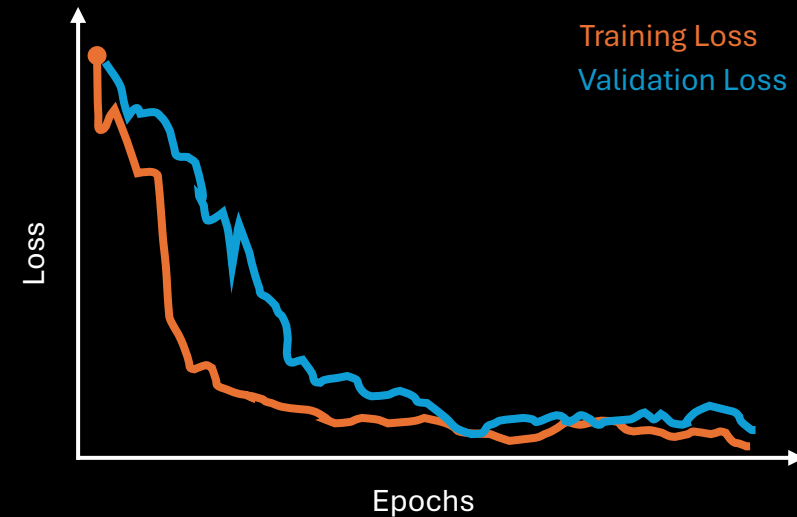
# Applying a trained network/model

- Typically called *Inference* or *Prediction*, is the application of a trained model (network) to new unseen data.



# How to follow network training?

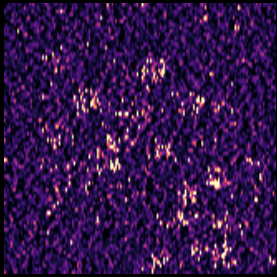
- Each forward and backward pass is a step
  - A step passes a batch of images
- An epoch is when the network sees all the training data
  - Many steps until all data is seen
- Training happens over many epochs
  - Training loss converges to zero (0)
- Split data into 3 groups:
  - Training: used to update and tune the network
  - Validation: test the model for unseen data during training
  - Test: new data to test model generalization properties
    - Foundational Models are the best generalizers, trained with huge amounts of data



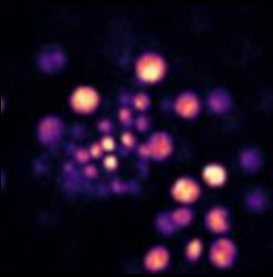
# Learning Strategies

## Supervised

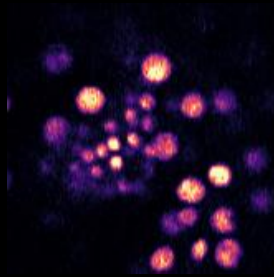
Compare



Input

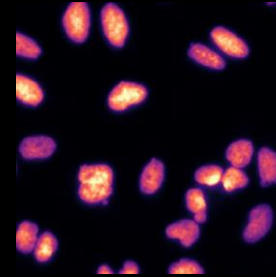


Predicted

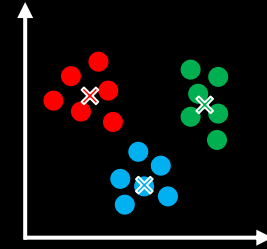


Ground Truth

## Unsupervised



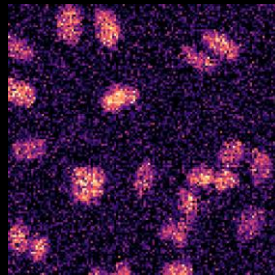
Input



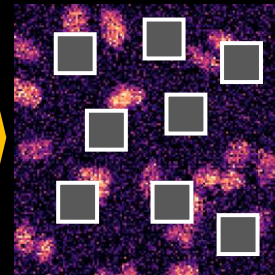
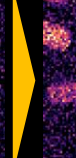
Predicted

## Self-Supervised

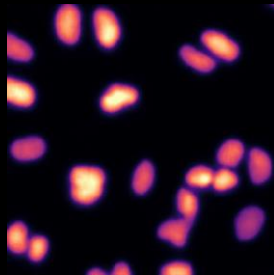
Compare



Input



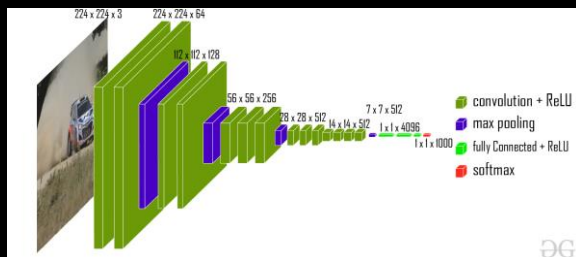
Degraded



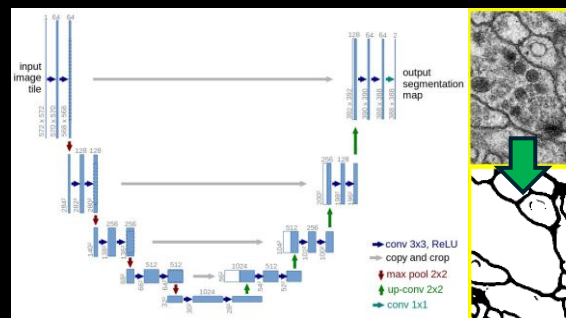
Predicted

# Common DL architectures and their use cases

VGG-16 (CNN)  
classification

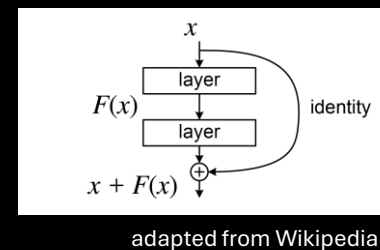


U-Net (CNN)  
image segmentation (Classification)



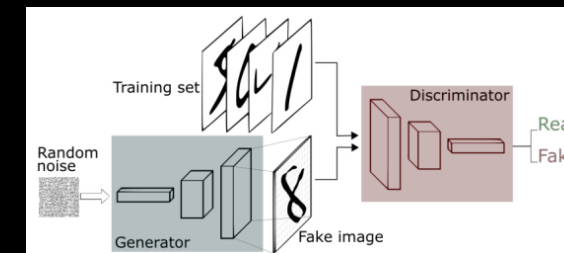
Ronneberger *et al.* 2016

ResNet  
image classification



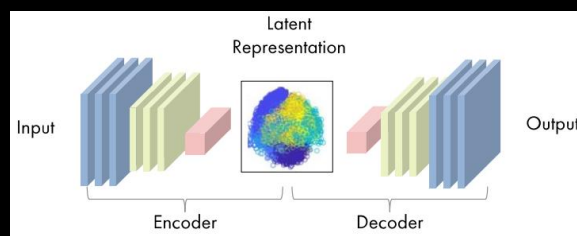
adapted from Wikipedia

Generative Adversarial Networks (GAN)  
image generation

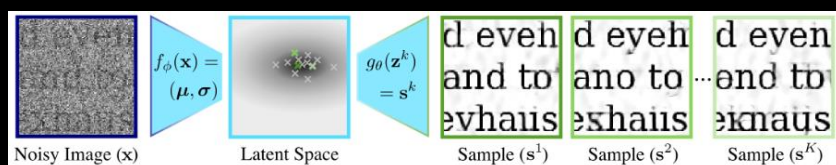


Schematic by Mariana Ferreira

Variational Auto-Encoder  
image generation

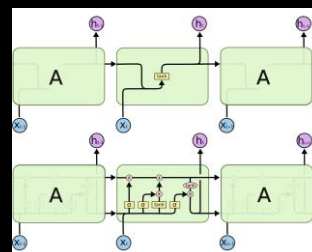


Schematic by Mariana Ferreira



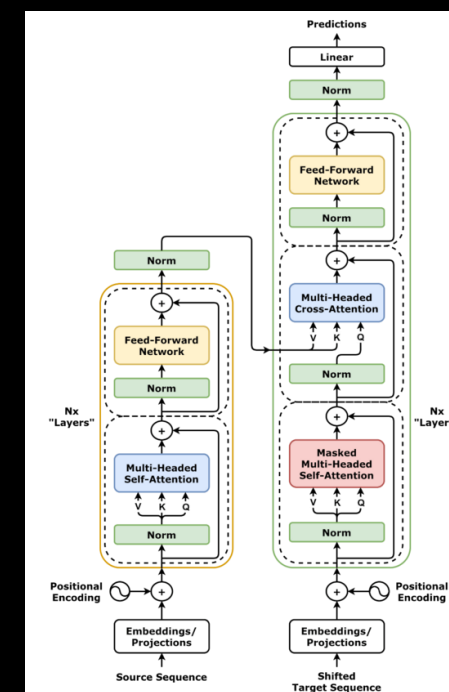
Prakash *et al.* 2021 DivNoising

Long short-term memory  
sequential data



Transformers  
text/sequential data

Attention is all you need. *Neurips* 2017

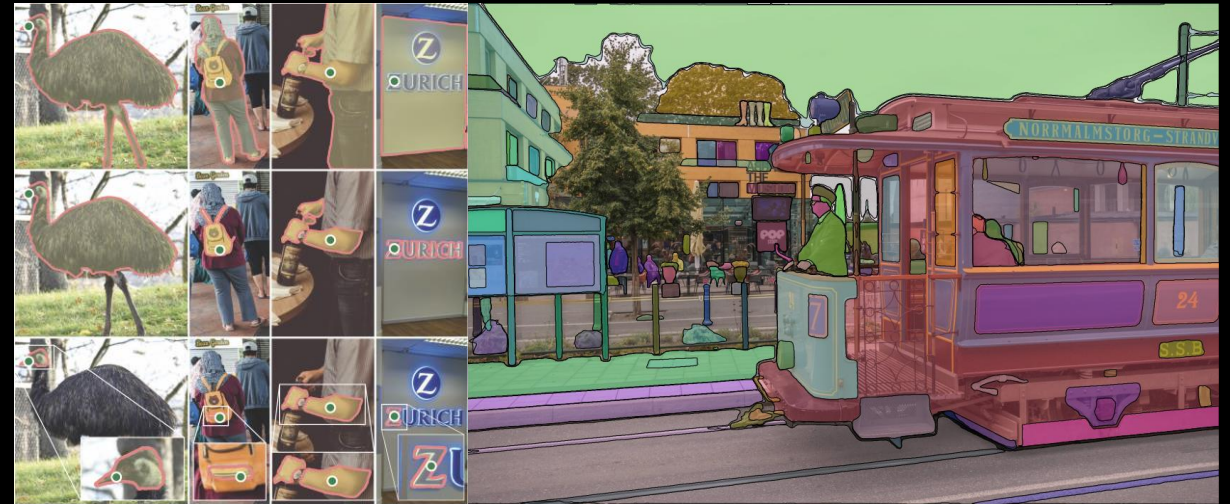




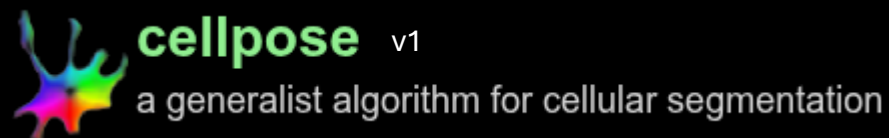
# Foundation models

- Large models trained on vast amounts of data (and some can perform general tasks).
  - Large Language Models (LLM) are a type of foundation models
- Google's BERT: baseline for natural language processing
- OpenAI's GPT
  - LLM: ChatGPT, Claude, Llama, Gemini, Grok, etc
- Image generation models:
  - Dall-E (OpenAI)
  - Stable Diffusion (Stability AI)
  - Imagen (Google)

## Segment Anything Model (SAM) from Meta AI



adapted from Meta AI (<https://github.com/facebookresearch/segment-anything>)



Cellpose-SAM